AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

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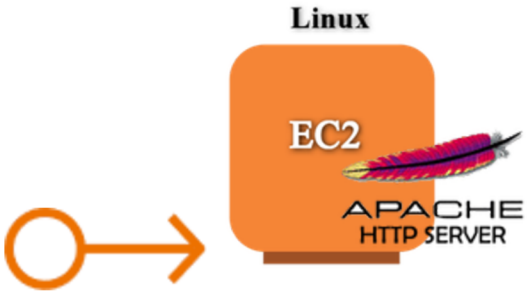
AWS Academy Certified Educator

6. Configure Elastic IP Address to Windows Web Server in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

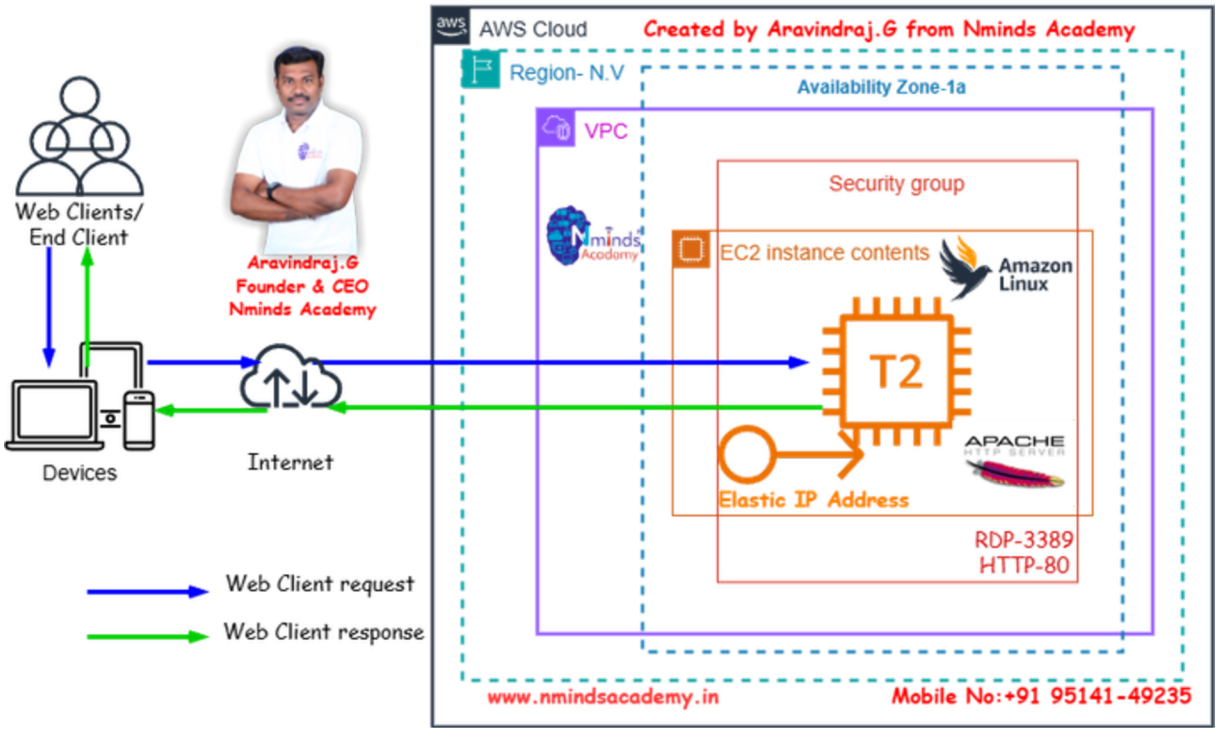
1. Highly Available Applications:
   - If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.
2. Web Servers:
   - If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.
3. Disaster Recovery:
   - Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

Best Practices:

- Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.

Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS

Used Services in Windows Server

Amazon EC2

APACHE HTTP SERVER

Apache Webserver

Amazon Elastic IP Address

```
root@ip-172-31-17-156:~

  Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64                                   13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                        1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                   2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                              3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                           4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                      5/13
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                                     6/13
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                7/13
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                          8/13
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                               9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                 10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                  11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                12/13
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                  13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                          apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64                  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
  httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                 libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch                     mod_http2-2.0.27-1.amzn2023.0.3.x86_64
  mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-17-156 ~]# systemctl start httpd
[root@ip-172-31-17-156 ~]# systemctl start httpd enable
Failed to start enable.service: Unit enable.service not found.
[root@ip-172-31-17-156 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
```

13.50.204.15

It works!